INTERNAL DEVELOPMENT REFERENCE

iSurvix Field Development Guide

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This internal guide records the approved company access model, review implementation, preservation rules and release evidence. It is maintained separately from the Software Manual so the final end-user release can contain operating instructions only.

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Baseline and release boundary

Project root: F:\iSurvix\iSurvix-Field. Keep runtime data under its .local folder. Local Django and Ollama are loopback-only. This version extends the existing photo/checklist pilot and company access; it does not establish production readiness or real-site AI accuracy.

Final release packaging must include the Software Manual only. Keep this guide, engineering plans, test logs, source references and credential inventories internal. Do not delete source, audit records or backups merely to simplify the user-facing documentation.

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Approved architecture and access policy

Company is the security boundary. Explicit project membership is the unit of access, and the account role remains the unit of authority.

Layer	Policy
Company	Groups related projects and affiliated people. Cross-company team grants are prohibited.
Project Admin	Can manage multiple explicitly granted projects within the same company. No implicit access to every company project.
Team Lead Engineer Client	One initial primary project; extra same-company projects require explicit administrator grants.
Working context	A scoped project selector focuses the workspace. A posted or queried project ID never grants authority.
Primary Admin	Project responsibility designation; replacing it does not revoke earlier explicit grants.
Compatibility	Existing primary profile project is preserved for legacy records; explicit membership/revocation controls current access.

Delivery responsibilities

Backend ownership covers company/membership schema, conservative backfill, governance, invitations, selection and scope queries. UX ownership covers project context, grants and private/shared review controls. AI ownership covers local structured review and disclosure gating. Root integration covers combined checks, deployment and the split documentation.

Planning assumption

“One project at a time” is implemented as a primary/default project and selected workspace context. Explicit additional grants allow the same person to work across permitted projects without duplicate identities. It does not prohibit retaining historical assignments or require deleting work when context changes.

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Schema and migration preservation

Use additive schema changes and conservative access backfill. Preserve users, credentials, answers, media, template snapshots, activity status and approvals.

Data	Implementation intent
Company	Company record and project/company association group existing projects without adding cross-project access.
ProjectMembership	Unique user/project grant with active state records additional access and explicit revocation.
Profile	Company affiliation and existing primary project retain role and compatibility.
Legacy migration	Existing primary access becomes an explicit membership. Neutral legacy company assignment does not create additional project rights.
Critical review	Persist checklist snapshot fingerprint, structured findings, request actor/model/provenance and sharing state.
Historical closure	Existing Final/completed activities and nullable Client routing remain unchanged by membership migration.

Change controls

Take a consistent database checkpoint before applying migrations. Serialize grant, invitation acceptance and review decisions through the existing governance transaction lock. Recheck the actor and target project under the lock. Do not infer access merely from a company label or a posted membership ID.

Compatibility audit

Update direct primary-project filters in governance, assignment forms, checklist access, photo endpoints, legacy scoped views, template management, text AI context and Client configuration. An explicit inactive grant must defeat legacy primary fallback. Grant revocation removes access without rewriting historical ownership.

Primary and additional grants

The initial invitation chooses the primary project. Additional grants must remain inside the same company and the manager’s authority. Super Admin handles permitted account provisioning, employer Company scope, industries, projects and administrative site records. Only industry Admin creates client organizations and invites or grants Client reviewer accounts. Project Admin handles operational administration within granted scope. Existing tasks and exported evidence retain their original project identifiers.

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Existing review workflow invariants

The multi-project change expands permitted project sets while preserving the meaning of every review signature.

Invariant	Required result
Engineer submission	Only the actual assigned Engineer can submit, including through legacy delegates.
Internal review	Assigned Quality Engineer confirms the submitted snapshot before an actual Team Lead with project access records internal approval or rejection. Historical signatures retain their recorded version.
Client decision	Only an authenticated Client can satisfy a required Client gate.
Administrative corrections	Project Admin/Lead support is limited to unlocked non-Final fields; it does not create stage-signing authority.
Final locks	Final engineering answers remain locked. Full rejection clears approvals only in eligible pre-Final states.
Client snapshot	First internal completion freezes Client requirement using effective memberships and valid pending invitations.
No-Client branch	Close after Team Lead completion only when the frozen Client requirement is false.
Client rejection	Leave activity open at Client review and keep engineering answers locked.
Stale requests	Recheck signed snapshot and current linked activity state before decisions.

Query scope

Use central accessible-project helpers. Engineer queries additionally filter assigned ownership. Client queries additionally require Final status. Administrator template and audit queries remain constrained to managed projects. No global queryset becomes safe simply because the visible project selector is scoped.

Invitation scope

Invitation acceptance is single-use, email-bound and issuer-authorized at acceptance time. Admin may invite Engineer, Team Lead, Quality Engineer or Client to a managed project; only Super Admin may invite a Project Admin. Issuer membership revocation must invalidate authority even when the invitation token itself is unexpired.

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Critical AI review and disclosure

The critical review is a separate advisory artifact for Team Lead assessment. It never becomes an automatic approval, checklist mutation or certification.

Input and inference

Use the frozen checklist schema, current answers and bounded evidence metadata. Treat all supplied answers and image-derived text as untrusted evidence, not instructions. Call the configured native local Ollama API through the shared inference lease. Unknown, unavailable or malformed responses fail safely. Photo analysis separately retains the model vision-capability check.

Critical checklist review currently runs as a synchronous, bounded batch request. It is separate from the durable photo queue. Large catalogs can take several model calls; sustained latency and recovery after a web request interruption still need acceptance testing.

Output validation

Validate structured findings, severity, uncertainty and known field or section identifiers. Preserve evidence references and missing-information questions. Do not invent readings, dimensions, standards clauses or compliance conclusions. Bounded inputs must reject unsupported size or explain omitted material rather than silently presenting an incomplete review as complete.

Visibility state	Required behavior
Internal default	Only permitted internal review roles may read the artifact; Client access is denied.
Shared current review	Actual Team Lead explicitly shares; Client additionally needs project access and Final status.
Unshared	Withdraw Client access immediately and audit the action.
Stale inputs	Changed answers/schema/evidence invalidate current sharing; request and assess a fresh review.
Generic export/context	Do not automatically include internal reviews in Client exports, text context, evidence reports or client decision tokens.

Audit and race controls

Record requesting and sharing actors and timestamps without credentials. Compare the snapshot after inference before persisting or sharing. Sharing must recheck current authority and source state, not merely trust a browser checkbox. Client approval remains independent of whether the review was shared.

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Verification and release acceptance

Record exact results after the implementation stabilizes. Earlier suites provide regression history, not proof that newly added behavior passed.

Area	Required verification
Migration	Existing account/project counts, answers, photos, approval history and completed activity state preserved; expected memberships backfilled only.
Company scope	Multi-project Admin positive cases; foreign-company denial; forged IDs; inactive grants and issuer revocation.
Role grants	Additional Team Lead, Engineer and Client access; Engineer ownership still enforced; revoked grant denies old primary access.
Workflow	Strict stage signatures, no-Client closure, Client required routing with extra grants, stale Client decisions and historical preservation.
AI review	Assigned Engineer or actual Team Lead requests; Team Lead alone shares. Structured fields, inference lease, stale snapshots and private/shared Client cases are checked.
Leakage	Client direct endpoint, generic Excel/JSON/text context and cross-project access do not disclose internal review.
Interface	Desktop/mobile grants, selected project, internal/share labels and read-only Client presentation.
Runtime	Apply migrations, Django check, migration consistency and authenticated read-only live checks.

Evidence baseline

The earlier company release passed 145 combined tests, including company/migration and critical-review checks. Subsequent workflow recovery passed 147 combined tests, followed by Engineer upload checks. These are dated regression baselines, not the final field-workflow acceptance count. Runs overlap and must not be summed as unique tests.

Python 3.12.10 and qwen3.5:4b completion/vision capabilities were verified locally. A real native Ollama critical-review probe returned one validated synthetic finding. It created no project records. Earlier native synthetic photo processing also passed; neither probe establishes real-site accuracy.

Migrations 0020 and 0021 were applied after a consistent database checkpoint. Every old column in the preserved user/profile/project/checklist/task/template/photo/draft/audit records matched its pre-upgrade hash, and all media matched. Existing counts were 16 users, 2 projects, 3 template revisions and 57 audit events; no site activities or photos were fabricated.

Open field and production acceptance

Real-site AI accuracy, GPU performance, sustained 400-photo throughput, load latency, email delivery, external hosting, production hardening and an independent restore drill remain unverified. Measured plan/elevation and editable SVG/DXF remain planned. Do not substitute a synthetic pass for field acceptance.

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Operations and distribution

Keep application data and deliverables in the approved workspace. The current release work is local; no hosted deployment is implied.

Local operation

Start-iSurvix.ps1 manages the loopback Django app and independent photo worker. The project Python environment is under .local. Database, original media, uploads, logs, secrets and checkpoints stay under the project's .local directory. Preserve original image bytes and checklist snapshots during upgrades.

Predeployment procedure

Back up the database consistently, stop only the verified launcher and its children, apply additive migrations and restart hidden with local logs. Check the migration list, Django configuration, worker health and authenticated project-scoped pages. Never use a broad process kill or a destructive reset as an upgrade step.

Source reference

Topic	Reference
Approved plan	docs/COMPANY_ACCESS_AND_AI_REVIEW_PLAN.md
Local brief	docs/LOCAL_CODEX_HANDOFF.md
Access and governance	backend/core/access.py; governance models, forms and views
Review and workflow	backend/core/workflow_service.py; template_views.py; checklist_ai_review.py
Evidence processing	backend/core/photo_service.py; photo_queue.py; management/commands/photo_worker.py
User guidance	docs/manual/iSurvix_Field_Software_Manual.docx and .pdf

Final release contents

The end-user documentation package contains only the Software Manual in editable Word and PDF formats. Exclude this Development Guide, superseded combined guide, implementation plans, private credentials, source snapshots, test databases and logs. Keep internal engineering evidence in the repository; removal from the release package does not mean deleting the development record.

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Field intake and evidence placement

The approved field extension combines bulk site preloads, photo-first execution, durable vision tagging, explicit placement confirmation and current critical review.

Component	Delivered implementation
0022 assignment intake	Additive intake_metadata and AssignmentIntakeBatch. CSV preview stores validated rows and source/template fingerprints; atomic commit rechecks creator, roles, memberships, duplicates and current published schema.
0023 photo locations	PhotoEvidence group plus PhotoPlacement links separate from answer_data/image_data. Model/job, schema fingerprint, field/section, uncertainty and review provenance are retained.
Durable analysis	Civil/Electrical/Tower uploads use bounded batches. Opt-in queued group jobs wait behind active work under the shared inference lease, with pending-job and duplicate-work bounds.
Human correction	Proposed, confirmed, manual and rejected states. Transactions recheck current scope, schema and every target approval lock. Human corrections survive re-analysis.
Coverage and export	Canonical master has no dedicated photo fields and 150 inspection items. Recommended evidence is separate from explicit required photos. Confirmed links render in checklist and Excel and satisfy matching photo requirements.
Final critique	Current photo processing readiness and mapping/coverage summaries feed the review. Engineer reads own requested reviews; Team Lead retains exclusive Client sharing. Corrections invalidate old snapshots.

Evidence and limits

Final field regression passed all 186 tests in 163.371 seconds, including five integration cases for confirmed/proposed/stale exports, required photos and partial approval. Native vision correctly returned the matching blue subject only after explicit visible-evidence safeguards were added. Migrations preserved old values in 29 existing tables and all four media files. Live Engineer and Team Lead checks passed without changing site evidence or decisions. See the local handoff for the checkpoint and runtime details.

This release remains a local loopback application. Physical phone access, offline upload persistence, real-site accuracy, 400-photo throughput and production operation require separate acceptance. Existing original files, answers, approval history, credentials and company boundaries must survive migration unchanged.

Roadmap pilot implementation

This update documents the delivered optional pilot rather than the complete product roadmap. Keep existing review mode and historical approvals intact. Consult docs/LOCAL_CODEX_HANDOFF.md for current checks, migration evidence and runtime acceptance.

Navigation scope and taxonomy

Workspace GET view values are my_work, projects, reviews, reports and administration. The industry filter uses dynamic catalog choices, initially including telecom and construction. Project scope is preserved across primary navigation; industry changes reset project selection to all. Querysets must remain server-scoped. Project.industry and Checklist.workflow_type distinguish telecom_survey, telecom_inspection and construction_inspection. Construction survey remains outside this pilot.

Independent authority and qualification

Checklist.governance_mode distinguishes legacy and independent review. Independent policies bind L2 reviewer and L3 approver through checklist.review_policy. Actual assigned Engineer prepares and submits; new policy version 2 assigns Quality Engineer to L2 and Team Lead to L3. Historical version 1 retains Team Lead L2 and Project Manager L3 signatures. Shared project membership, administrative support and AI output do not confer stage authority. A permitted Team Lead records technical qualification against the exact template definition and workflow before independent submission; historical Project Manager qualification records remain preserved.

Versioned records and release gates

SubmissionRevision freezes source schema, recorded answers, evidence manifest and digest. Review decisions bind the exact revision and stage. Qualification captures criteria references, method and evidence expectations. InspectionFinding retains observation, remedy and independent retest events. Open blocking or critical findings prevent release. Preserve signature, approval and finding history; do not map old approvals into new signatures. Internal approval and configured Client acceptance remain distinct.

Source references

Review backend/core/independent_workflow.py, roadmap_service.py, roadmap_views.py, test_independent_workflow.py, test_roadmap_acceptance.py and test_roadmap_controls.py with current additive migrations and central access helpers. UI Site context must use guarded nullable reviewers and UTF-8 templates. Test routes through authenticated scoped HTTP requests, including newly created projects without an administrator.

Metadata processing and acceptance limits

The photo worker extracts reusable visual observations once per valid photo and configured model, then performs checklist matching. The same original evidence remains available to the Engineer, reviewer and report without treating AI observations as verified answers.

Capacity and queue semantics

Current UI and service capacity is up to 400 photos per site, with uploads in batches of 20. CSV assignment intake remains capped at 200 rows; do not increase it when changing photo limits. Group enqueue uses queue_when_busy to persist intent behind active work. Shared inference leasing serializes model work. Metadata readiness, analyzed-photo readiness, mapping review and coverage are separate state measures. Preserve human placement corrections and approval locks during reruns.

Reuse and progress contract

State exposes metadata_ready_count, per-photo metadata_ready and current visual_metadata. Job progress identifies extracting_metadata, matching_checklist, completed or failed, with photo_id and cache_reused. Show cached observations as unverified. A historical failed job whose photo_ids are all in current analyzed_photo_ids remains auditable without offering a redundant retry. Cache validity depends on the configured model and source content.

Verification and operational boundaries

See the local handoff for current suite counts and exact acceptance evidence. Synthetic UI coverage and disposable database tests do not establish real-site AI accuracy. The measured 78-photo run does not prove 400-photo throughput or a 30 minute service level. External hosting, physical phone connectivity, offline persistence, standards certification and the remaining roadmap modules require separate acceptance. Keep database, originals, logs and credentials under the approved workspace.

Release separation

Distribute only the end-user Software Manual DOCX and PDF in the user documentation package. Keep this Development Guide, source roadmap, tests, logs, credentials and implementation plans internal. Do not package the roadmap source PDF as an end-user manual. Retain the underlying development evidence locally.

Expiring grants and correction lineage

Migration 0028 adds optional ProjectMembership.expires_at. Server authorization denies expired memberships, including legacy primary fallback. In independent mode a Client rejection creates an immutable C1 decision and a linked editable correction activity requiring a fresh L1 to L2 to L3 cycle. Preserve the approved source, historical Client record and frozen reports. Legacy Client rejection behavior remains unchanged. Verify expiry boundaries, correction ownership, revision lineage and cross-project denial in the current handoff checks.

Super Admin console and authority

The Super Admin console is administration only. Its global navigation is Dashboard, Industries, Clients, Projects and Sites. Legacy Super workspace view names map to the administration dashboard. Other roles retain operational navigation and server scoped project access.

Catalog and organization model

IndustryCatalog stores code, name and active status. Referenced codes remain permanent; deactivation preserves history. Project industry selection uses catalog values rather than a fixed two industry list. A newly configured industry does not create a supported engineering workflow. Clients are company records and must remain distinct from Client reviewer identities.

Authorization boundary

Super Admin provisions permitted non Client access and configures employer Company scope, industries, projects and Project Admin ownership. Super Clients is read only; Admin owns client organization creation and Client reviewer provisioning. Administrative site records expose identifiers, assignment ownership and status only. Deny engineering pages and mutations, assignment execution, technical template operations, photo or AI actions and technical or Client signatures. Keep operational administration with scoped Project Admin and stage signatures with the actual assigned actors. Hiding a link is not authorization; direct endpoints must enforce the same boundary.

Current verification reference

Consult docs/LOCAL_CODEX_HANDOFF.md for current full suite and live read only acceptance evidence. Verify exact console navigation, dynamic third industry creation, inactive catalog behavior, preserved historical references, unassigned project administration, administrative site scope and direct engineering endpoint denial. Do not infer new workflow support from a new catalog code.

Industry administration and quality review

This update adds industry bound Admin scope and Quality Engineer review while preserving existing users, company boundaries, evidence, approval locks and frozen revision signatures. Consult the local handoff for current test counts and operational evidence.

Industry binding and role provisioning

Profile.admin_industry references IndustryCatalog. Only Super Admin may set or change an Admin binding. An ADMIN invitation requires active industry selection consistent with its initial project. Admin scope intersects industry, company and explicit project permissions. Admin creates ClientOrganization records and provisions Client reviewer accounts and project roles. Super Admin cannot create clients, invite Client reviewers or grant new Client access. QUALITY_ENGINEER is an explicit role, not an administrative privilege.

Current quality review contracts

Legacy activities use Checklist.quality_reviewer and snapshot bound quality_approval before Team Lead approval. Quality review POST binds expected_updated_at and decision approve or reject; rejection requires reason. Existing eligible activities may assign an active project Quality Engineer with reason and stale check while preserving answers, original evidence and section approvals. Relevant source changes require revalidation.

Versioned independent policies

New independent policy and submission workflow_version 2 maps l2_reviewer to Quality Engineer and l3_approver to Team Lead. Version 1 frozen revisions retain Team Lead and Project Manager semantics. Display authoritative quality_flow derived from the current revision, not merely the latest policy. Eligible Admin draft or returned policy upgrades choose QE, TL and optional named Client with reason and expected timestamp. Existing signed source records are retained; new submissions carry the new policy.

Authorization and regression checks

Verify industry self change denial, mismatched initial project, foreign or inactive reviewers, strict Engineer ownership, QE before TL ordering, stale snapshots, rejection reasons, frozen v1 report fidelity, returned activity upgrade and Client correction lineage. Test form selectors and direct endpoints. Do not use an old successful suite as evidence that current changes passed; current results are recorded in docs/LOCAL_CODEX_HANDOFF.md.

Client organizations and ownership

Client ownership is an additive domain correction. Employer Company tenancy remains unchanged. Do not import or rename historical Company rows into new client records automatically.

Model and scope

ClientOrganization records name, description, employer company, industry and creating actor. Project.client_organization is an optional link. The Admin creation form accepts name and description; server derives company and industry from the bound Admin. Project assignment rechecks managed project access, tenant identity and matching industry.

Role boundaries

Only Admin creates or assigns client organizations and invites or grants Client reviewer accounts. Super Clients is a read only organization list. InvitationForm excludes CLIENT for Super while retaining it for Admin. Direct service and HTTP endpoints enforce the same restriction; a removed browser option alone is insufficient.

Interface and compatibility

Super Dashboard owns permitted account provisioning. Projects uses Company labels for employer tenancy. Admin Administration offers client organization creation and project linking, separate from Client person invitations. Preserve old Company/project identities, answers, evidence, review snapshots, grants and historical audit records. No client creation implies new workflow capability or a technical signature.

Verification reference

Check current suite results in docs/LOCAL_CODEX_HANDOFF.md. Cover Super denied create/invite/grant requests, Admin positive cases, mismatched tenant or industry, inactive/revoked scope, optional project linkage and unchanged existing records. Verify desktop and phone visibility plus actual form payloads using synthetic data.